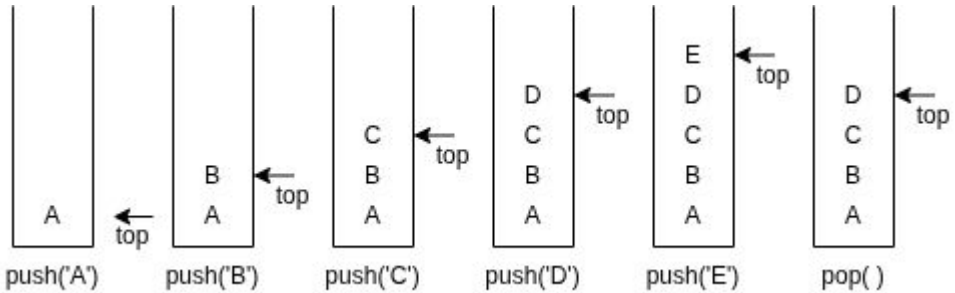
Lecture 14

Stack

- A stack is an Linear data structure in which insertions (also called pushes & adds) and deletions (also called pops & removes) are made at one end called the top.
- Given a stack S = (a0, a1, …, an-1), we say a0 is the bottom element and an-1 is the top element, ai is on top of element ai-1, 0<i<n.
- The restrictions on the stack imply that if we add the elements A, B, C, D, E to the stack in that order then E is the first element we delete from the stack.
- Since the last element inserted into is the first element removed, a stack is also known as Last-In-First-Out (LIFO) list.

Inserting and deleting elements in a stack

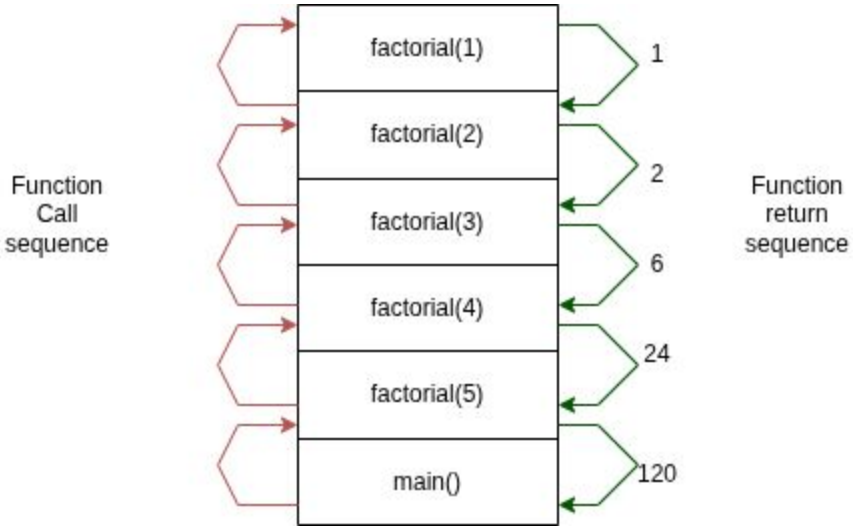


example

Recursive functions are implemented using a stack.

- Finding the factorial of a given number.
- ```
  int factorial(int num){
      if(num == 0 || numb == 1)
      return 1;
      return num * factorial(num - 1);
   }
  ```
- Calling the factorial function , factorial(5) will return 120.
  ```

continue…


Function
Call
sequence
factorial(1)
factorial(2)
factorial(3)
factorial(4)
factorial(5)
main()
1
2
6
24
120
Function
return
sequence

Stack Creation

```
#define MAX_STACK_SIZE 5
typedef struct{
     int key;
}element;
element stack[MAX_STACK_SIZE];
int top = -1;
```

```
#define MAX_STACK_SIZE 5
int stack[MAX_STACK_SIZE];
int top = -1;
```

Stack Operations

1. Push
2. Pop
3. isFull
4. isEmpty

Push

```
push(element):
    /* check for stack overflow */
    if(top == size - 1):
        stack is full
        return
    top++;
    stack[top] = element
```

Pop

```
pop():
    /* check for stack underflow */
    if(top == -1):
        print stack is underflow
    char ch;
    ch = stack[top]
    top--
    return removed element
```

isFull

```
isFull():
    if(top == size - 1):
        return true
    return false
```

isEmpty

```
isEmpty():
    if(top == -1):
        return true;
    return false;
```

Problem

Given the elements present in the stack pop the middle element. The size of stack is 5. The first stack in the given picture is input and the second stack is output.